

```
1  #include <stdio.h>
2  int main()
3  {
4      int numberOfProcesses;
5      printf("Enter number of processes: ");
6      scanf("%d", &numberOfProcesses);
7
8      int processId[10];
9      int burstTime[10];
10     int priority[10];
11
12     // Input process details
13     for(int i = 0; i < numberOfProcesses; i++)
14     {
15         processId[i] = i + 1;
16
17         printf("Enter burst time of P%d: ", processId[i]);
18         scanf("%d", &burstTime[i]);
19
20         printf("Enter priority of P%d: ", processId[i]);
21         scanf("%d", &priority[i]);
22     }
23
24     // Sort processes based on priority
25     for(int i = 0; i < numberOfProcesses - 1; i++)
26     {
27         for(int j = i + 1; j < numberOfProcesses; j++)
28         {
29             if(priority[j] < priority[i])
30             {
31                 int temp;
32
33                 temp = priority[i];
34                 priority[i] = priority[j];
35                 priority[j] = temp;
36
37                 temp = burstTime[i];
38                 burstTime[i] = burstTime[j];
39                 burstTime[j] = temp;
40
41                 temp = processId[i];
42                 processId[i] = processId[j];
43                 processId[j] = temp;
44             }
45         }
46     }
47
48     int waitingTime[10];
49     int turnaroundTime[10];
50
51     // Calculate waiting and turnaround time
52     for(int i = 0; i < numberOfProcesses; i++)
53     {
54         if(i == 0)
55             waitingTime[i] = 0;
56         else
57             waitingTime[i] = waitingTime[i - 1] + burstTime[i - 1];
58
59         turnaroundTime[i] = waitingTime[i] + burstTime[i];
60     }
61
62     // Output
63     printf("\nPID\tBT\tPR\tWT\tTAT\n");
64     for(int i = 0; i < numberOfProcesses; i++)
65     {
66         printf("P%d\t%d\t%d\t%d\t%d\n",
67             processId[i],
68             burstTime[i],
69             priority[i],
70             waitingTime[i],
71             turnaroundTime[i]);
72     }
73     return 0;
74 }
75
```